F1 Tenth Final Project Proposal

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Team 10 — March 2026

1 Abstract

Depth estimation is fairly important in tasks like SLAM and autonomous racing, yet achieving it cost-effectively remains a significant challenge. High-density LiDAR sensors provide accurate metric depth but are expensive for consumer-scale applications, while pure monocular methods produce scale-ambiguous relative depth. We aim to propose a novel, training-free approach to metric depth estimation by fusing a monocular camera with a sparse, low-cost 1D LiDAR beam array. We aim to "anchor" the scale of a monocular depth model using a set of accurate distance measurements. To handle the noise in the real world, our approach will leverage RANSAC to actively reject outlier and degraded measurements. We will evaluate our method on our F1Tenth-scale vehicles using a standard camera and a subset of measurements from the provided LiDAR. We will quantify our result through common metrics measuring depth like Absolute Relative Error (AbsRel) and robustness to challenging scenarios like dynamic obstacles. This work seeks to demonstrate that accurate metric depth can be achieved using commodity hardware and is model agnostic, bridging the gap between high-cost sensors and unreliable monocular vision.

2 Introduction

The F1Tenth competition provides an ideal platform for exploring the intersection of autonomy, performance, and cost. For a race car, depth perception is not just for navigation but for precise localization, obstacle avoidance, and trajectory optimization at the limits of handling. Current state-of-the-art solutions for depth perception, such as high-resolution 3D LiDAR, offer the dense, accurate data required for such tasks but represent a prohibitive cost barrier for mass deployment. Conversely, pure monocular depth estimation, while inexpensive, suffers from a fundamental ambiguity: the projection of a 3D world onto a 2D image loses the scale of the scene, providing only relative depth information.

This project addresses the critical need for a middle ground: a solution that combines the affordability of a camera with the metric accuracy of a LiDAR. The core idea is to use a simple, low-cost 1D LiDAR array (e.g., a single-beam sensor) to provide a sparse but accurate "anchor" for the dense, but scale-ambiguous, depth map produced by a monocular network. This concept is particularly relevant for F1Tenth, where the vehicle is small, fast, and operates in a structured environment. By leveraging existing pre-trained monocular depth models in a training-free manner, we can focus on the core challenge of fusing these two disparate data streams. Success in this project would demonstrate a pathway to robust, low-cost depth perception, making autonomous racing and other robotics applications more accessible. We hypothesize that a single accurate anchor point, when intelligently fused, is sufficient to recover global scale and improve the accuracy of monocular depth estimation

3 Related Works

Our work combines monocular depth estimation, sensor fusion, and robust statistics.

- **Sensor Fusion for Depth**: Traditional approaches to metric depth often rely on stereo vision [3] or Structure-from-Motion (SfM) [4]. However, stereo systems often fail in texture-less environments or require significant computational resources. SfM, while powerful, relies on feature tracking over time, which is challenging for high-speed racing where motion blur and rapid viewpoint changes are common. Sensor fusion methods, such as those combining sparse LiDAR with images (e.g., [5]), typically require dense depth maps or complex, often learned, fusion architectures.
- **Monocular Depth Estimation**: Recent work has focused on learning-based depth estimation. Methods like MiDaS [1] and Depth Anything [2] excel at predicting relative depth from a single image by leveraging robust backbones for semantic extraction. However, we argue that these models only provide depth up to an unknown scale and shift, making them unsuitable for direct use in metric tasks like racing. In contrast to purely monocular approaches, we leverage powerful learning-based models for their dense, high-quality relative depth maps while correcting their metric ambiguity using external sensors. Our work differs by using an extremely sparse LiDAR to correct a pre-existing monocular model without any retraining, focusing on algorithmic robustness to sensor noise and outliers.
- **Robust Estimation & Outlier Rejection**: RANSAC (Random Sample Consensus) [6] is a classic algorithm for estimating model parameters in the presence of a large proportion of outliers. In the context of sensor fusion, it can be used to find a transformation (e.g., scaling factor) that is consistent across multiple "inlier" points, while ignoring erroneous correspondences. We propose to use RANSAC to robustly identify valid LiDAR-to-pixel correspondences, ensuring that a single erroneous measurement from a moving object or reflective surface does not corrupt the entire depth map. This is a more principled approach than simple averaging or Kalman filtering when a subset of measurements may be unreliable.

4 Approach

Our proposed method is a three-stage, training-free pipeline for metric depth estimation.

4.1 System Setup

We will equip an F1Tenth vehicle with a standard monocular camera and a 1D LiDAR (we plan to do it by using a subset of the current mounted lidar).

The LiDAR will be calibrated to the camera's coordinate frame, allowing us to project its 3D measurement onto the image plane to identify its corresponding pixel.

4.2 Pipeline

The pipeline for depth estimation consists of the following steps:

1. **Dense Relative Depth**: The monocular camera captures an image, which is passed through a pre-trained monocular depth model. The model outputs a dense, scale-ambiguous depth map, $\mathcal{D}_{rel}$.

2. **Sparse Absolute Anchor**: Simultaneously, the 1D LiDAR records an array of metric depth measurements, d_{lidar}. Using known extrinsic and intrinsic calibration, we project the 3D points onto the image to find its pixel location, p. This gives us "anchor" points: we know the ground-truth metric depth d_{lidar} for pixel p.

3. **Robust Scale Recovery with RANSAC**: The goal is to find a global scaling factor s such that $s * \mathcal{D}_{rel}(p) \approx d_{lidar}$. However, our anchor point might be invalid due to sensor noise, occlusions, or dynamic objects. To handle this, we will use a RANSAC-like framework.

    - **Idea**: Rather than relying on a single anchor, we leverage the full array of LiDAR beams, each providing a candidate anchor point $(\mathcal{D}_{rel}(p_i), d_{lidar,i})$.
    - Some of these measurements may be corrupted by sensor noise, reflective surfaces, or dynamic objects. RANSAC iteratively samples random subsets of anchor points to solve for s (and optionally a bias term), then counts how many remaining points agree with this hypothesis within a tolerance.

5 Evaluation

We will evaluate our system on the F1Tenth platform, using a high-precision ground truth provided by 3D lidar scan borrowed from MRSD program or manually increase the level of the attached lidar for quantitative analysis. We will align the lidar and camera data using timestamps provided in the rosbag recordings.

- **Dataset Collection**: We will collect synchronized camera and LiDAR data under various racing scenarios, including:
    - **Static Scenes**: The vehicle stationary with static obstacles.
    - **Dynamic Scenes**: An object or another car moving through the sensor's field of view.

  - **Challenging Surfaces**: Driving near reflective or texture-less surfaces.

- **Metrics**: We will use standard depth estimation metrics to quantify performance:
  - **Absolute Relative Error (AbsRel)**: $\frac{1}{n}\sum\frac{|d_i - d_i^*|}{d_i^*}$. This is our primary metric, measuring the overall scale accuracy.
  - **Mean Absolute Error (MAE)**: $\frac{1}{n}\sum|d_i - d_i^*|$. This provides a direct sense of error in meters.
  - **Root Mean Squared Error (RMSE)**: $\sqrt{\frac{1}{n}\sum(d_i - d_i^*)^2}$. This will penalize large errors, such as those from scale ambiguity or outlier corruption.

- **Experiments**:
  1. **Baseline Comparison**: We will compare our final metric depth map to a "naive" fusion (e.g., simply scaling $\mathcal{D}_{rel}$ to match a single, unfiltered LiDAR reading). This will quantify the benefit of our robust RANSAC-based approach.
  2. **Robustness Test**: We will introduce a dynamic object into the scene and evaluate the stability of the scale factor s^*. We expect our RANSAC method to maintain a consistent scale, while the baseline method will fail catastrophically when the LiDAR measurement comes from the moving object.

6 Timeline

- **Week 1 - Simulation Pipeline**: Benchmark Depth Anything V2 variants (Small/Large/Metric). Establish ground truth comparison framework.
- **Week 2 - 1D LiDAR Fusion**: Implement recovery using sparse 1D LiDAR anchor points. Evaluate metric accuracy vs. ground truth 2D scans.
- **Week 3 - Integration & Midstream**: Connect metric depth output to navigation stack (gap finder / pure pursuit). Present midstream results. Begin hardware deployment.
- **Week 4 - Hardware & Racing**: Deploy on physical F1Tenth car. Tune for real-time performance. Test in NSH Atrium. Race 3 preparation.
- **Week 5 - Demo & Report**: Final demo with live car demonstration. Write and submit report.

7 References

References

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